Unit 22: Application Development

Assignment Brief

Programme Title	**Pearson BTEC Level 5 Higher National Diploma in Computing**
Student Name/ID Number	
Unit Number and Title	**Unit 22: Application Development (Y/618/7436)**
Academic Year	2026
Unit Tutor	Syazwi Sazeli
Assignment Title	**Develop Real-world Web Application**
Issue Date	2 June 2026
Submission Date	All Activities – 18 September 2026

Submission Format

You should submit a zip file containing:

- Complete Declaration Form with Signature – Rename as: **BTECID - DECLARATION FORM**
- Individual written report (MS Word document) – Rename as: **BTECID - REPORT**
- Complete Project folder – Rename as: **BTECID – PROJECT FILE**
- Project Video – Rename as: **BTECID – PROJECT VIDEO**

You must rename your zip file as **BTECID - ASSIGNMENT**. Please note that assignments submitted by other names will not be marked.

Report should be written in formal business style using single spacing and font size 12. Include a table of contents, appropriate headings, and Harvard-style references where required.

Note on Plagiarism: Students must submit their own work. Any copied, paraphrased, or AI-assisted content must be properly cited and referenced using the Harvard referencing system. Failure to acknowledge sources may be treated as academic misconduct.

Unit Learning Outcomes
LO1 Produce a software design document for a business-related problem based on requirements LO2 Research design and development tools and methodologies for the creation of a business application LO3 Plan and produce a functional business application with support documentation LO4 Evaluate the performance of a business application against its software design document and initial requirements.
Transferable skills and competencies developed
Computing-related cognitive skills • Computational thinking (including its relevance to everyday life) • Understanding of the scientific method and its applications to problem solving • Demonstrate knowledge and understanding of essential facts, concepts, principles and theories relating to computing and computer applications • Modelling – use such knowledge and understanding in the modelling and design of computer-based systems for the purposes of comprehension, communication, prediction and the understanding of trade-offs • Recognise and analyse criteria and specifications appropriate to specific problems, and plan strategies for their solutions • Critical evaluation and testing – analyse the extent to which a computer-based system meets the criteria defined for its current use and future development **Computing-related practical skills** • The ability to specify, design and construct reliable, secure and usable computer-based systems • The ability to evaluate systems in terms of quality attributes and possible trade-offs presented within the given problem • The ability to plan and manage projects to deliver computing systems within constraints of requirements, timescale and budget • The ability to deploy effectively the tools used for the construction and documentation of computer applications, with particular emphasis on understanding the whole process involved in the effective deployment of computers to solve practical problems • The ability to critically evaluate and analyse complex problems, including those with incomplete information, and devise appropriate solutions, within the constraints of a budget

Vocational scenario
You and your team (2–4 members) are the founding developers of an early-stage startup that has been selected to pitch at **GameForge** — a regional hackathon competition dedicated to finding the next generation of impactful web-based games. **The Competition Theme** You have been invited to participate in a hackathon competition with one simple but challenging theme: “Build a game that solves a real problem.” Your task is to identify a genuine problem faced by a specific group of people — whether it is students struggling to revise, communities lacking engagement, employees needing training, or anything else your team believes in — and solve it through the medium of a web-based game. Your game does not need to be complex, but it must be meaningful. The judging panel, made up of investors, educators, and industry professionals, will evaluate not just whether your game works, but whether it genuinely addresses a real need, is well-engineered, and has the potential to grow. **Your Deliverables** As a startup team, you are required to produce the following deliverables: • A Software Design Document (SDD) that defines your problem, your users, your system, and your risks — before a single line of code is written • A fully functional web-based game connected to a database, with user authentication, role-based access, and at least one dashboard or report • A peer review session where another team plays and critiques your game • A final evaluation presentation where you defend your design decisions, reflect on feedback, and propose what version 2.0 would look like **Key Requirements (technology)** What game you build is entirely up to your team. The only requirement is that it runs in a browser, solves a real problem, and meets the technical and functional criteria below. Past teams have built quiz battle platforms for exam revision, typing racers to improve digital literacy, escape rooms for employee onboarding, and prediction leagues for campus events. The best ideas usually come from problems your team has personally experienced.

- Backend: PHP or Python or Java (Node.js optional for middleware/API glue)
- Database: PostgreSQL
- Query language: SQL / PL/pgSQL
- Runtime: XAMPP (or equivalent local stack for chosen language)
- Frontend: HTML/React + Bootstrap (or another CSS framework)

Data to Store (adapt to your game scenario)

- Core game entities (e.g., questions & scores; rooms & players; levels & progress; challenges & results)
- Users and role-based access control (RBAC)
- Activity/audit logs and basic analytics/metrics
- Feedback/peer review notes (for evaluation)

Mandatory Functionalities

- CRUD for core entities with input validation and robust error handling
- Authentication & RBAC (e.g., player/moderator/admin; player/game-host/admin; member/manager/admin)
- Search & filter for key records
- At least one dashboard/report (e.g., leaderboard, game history, player stats, score trends)
- Clean, responsive UI (Bootstrap or equivalent)

Collaboration & Best Practices

- Use GitHub (or equivalent) for version control, issues, pull requests, and milestone tracking
- Write clear README/setup steps; document architectural decisions and change logs
- Adopt a simple, stated development methodology (e.g., Agile/Iterative)

Creative Flexibility (optional, encouraged)

- Real-time features (live score updates, multiplayer rooms), sound effects, animations
- Achievements and badges, streak tracking, shareable result cards, multilingual UI
- Admin game builder (custom question sets, level editor), CSV score export, scheduled tournaments

Acceptance Criteria (applies to all game types)

- Clear problem statement, target user group, and game concept that addresses the identified problem
- Implemented CRUD + RBAC + validation + error handling on core entities
- Persistent data in PostgreSQL with meaningful reporting
- Evidence of peer testing/feedback and final evaluation tied back to SDD & requirements

Remember: at GameForge, a polished, well-documented game that solves a small problem convincingly will always beat an ambitious game that is half-finished and poorly justified. Start with the problem. The game will follow.

Assignment Activity and Guidance

<u>**Activity 1**</u>

You are tasked to do a presentation that explores how the needs of your project can be met, through the development of a computer application. Your presentation should include the following: -

- Analyse your chosen business-related problem using appropriate methods to produce a well-structured software design document
- Produce a well-defined problem definition statement for a business problem
- Support the problem definition statement with:
  - A set of user requirements
  - A set of system requirements
- Review areas of risk related to the successful development of a proposed application.

Activity 2

Write a report to research design and development tools and methodologies for the creation of a business application. For the development of a proposed application, you will:

- Research the use of a range of software development.
    - Tools
    - Techniques
    - Methodologies
- Justify the software development tools, and development methodology selected.

Finally, you will evaluate the solution to a business-related problem and the preferred software development methodology by comparing the various software development tools and techniques researched.

You will support your review with multiple real-world case study examples of established software development methodologies, tools and techniques where possible, and you will provide a list of well-researched sources to support any findings and conclusions. You will make sure that any research sources have been either peer-reviewed or verified by multiple sources or citations.

Activity 3

You will plan and produce a functional business application your chosen application with support documentation. You will:

- Conduct a peer review of:
    - The problem definition statement
    - Proposed solution
    - Development strategy
- Document any feedback given
- Interpret peer-review feedback and identify opportunities not previously considered
- Develop:
    - A functional business application based on a specified business problem
    - Support documentation for the business application
- Include supportive evidence of using the preferred tools, techniques and methodologies

Activity 4

Prepare a formal presentation evaluating your business application by comparing it to the software design document and initial requirements.

Your presentation should include:

Performance Review:

- How well the application addresses the problem statement
- How fully it meets the initial requirements

Critical Review of Development Stages (including risks):

- Design
- Development
- Testing

Peer Feedback:

- Present findings and collect feedback
- Document and reflect on peer input

Finally, you will justify improvements to the business application system made because of feedback and any feedback that was not acted upon, including opportunities for improvement and further development.

Recommended Resources

https://www.w3schools.com/

https://stackoverflow.com/

https://chat.openai.com/chat

https://course.ccs.neu.edu/cs5500sp17/process.pdf

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Produce a software design document for a business-related problem based on requirements		
P1 Produce a well-defined problem definition statement, supported by a set of user and system requirements for a business problem. **P2** Review areas of risk related to the successful development of a proposed application.	**M1** Analyze a business-related problem using appropriate methods to produce a well-structured software design document.	**D1** Evaluate the solution to a business-related problem and the preferred software development methodology by comparing the various software development tools and techniques researched.
LO2 Research design and development tools and methodologies for the creation of a business application		
P3 Research the use of software development tools and techniques for the development of a proposed application.	**M2** Justify the software development tools and development methodology selected.	

Pass	Merit	Distinction
LO3 Plan and produce a functional business application with support documentation		**D2** Justify improvements to the business application system made because of feedback and feedback which was not acted upon, including opportunities for improvement and further development.
P4 Create a formal presentation that effectively reviews of your business application, problem definition statement, proposed solution, and development strategy. Use this presentation as part of a peer-review and document any feedback given. **P5** Develop a functional business application with support documentation based on a specified business problem.	**M3** Interpret peer-review feedback and identify opportunities not previously considered. **M4** Develop a functional business application based on a specific software design document, with supportive evidence of using the preferred tools, techniques and methodologies.	
LO4 Evaluate the performance of a business application against its software design document and initial requirements		
P6 Review the performance of the business application against the problem definition statement and initial requirements.	**M5** Critically review the design, development and testing stages of the application development process including risks.	